Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Mains Electricity and the National Grid

Where the electricity in the wall comes from, how it reaches you, and how the plug keeps you safe.

Background Knowledge

Read this as you come in. Highlight the key words, and use the space to add your own notes.

A. Ohm's Law

Ohm's law: for a resistor at a constant temperature, the current through it is **directly proportional** to the potential difference (voltage) across it. Double the potential difference and the current doubles.

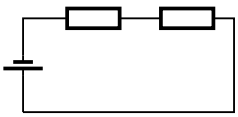
$$\boxed{V = I\,R}$$ potential difference (voltage) = current × resistance

Quantity	What it means	Unit, and how measured
Current, I	the rate of flow of **charge**: how much charge passes a point each second. $Q = I\,t$ (charge = current × time)	amps (A), an **ammeter** in series
Potential difference (voltage), V	the **energy transferred per unit of charge**; the push that drives the current. $E = Q\,V$ (energy = charge × p.d.)	volts (V), a **voltmeter** in parallel
Resistance, R	how much a component **opposes** the current. For the same potential difference (voltage), more resistance means less current.	ohms (Ω), found from $R = V \div I$

A component that obeys Ohm's law is **ohmic**: a straight-line I–V graph through the origin. A filament lamp is not: as it heats up, its resistance rises.

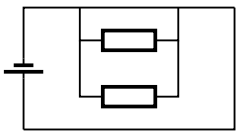
B. Series and parallel circuits

Series: components in **one loop**, one after another.



- The **current is the same** everywhere.
- The potential difference (voltage) of the supply is **shared** between the components.
- Resistances **add**: $R_{\text{total}} = R_1 + R_2$

Parallel: components on **separate branches**.



- Each branch gets the **full** potential difference (voltage) of the supply.
- The current **splits**; the branch currents add up to the total.
- Adding a branch **lowers** the total resistance.

Your home is wired in parallel. Every socket gets the full mains potential difference (voltage), and each appliance can be switched on and off by itself.

C. Power: electricity in everyday things

Power is the **rate of energy transfer**: how much energy is transferred each second. 1 watt (W) = 1 joule per second.

$P = V\,I$ power = potential difference (voltage) × current

$P = I^2\,R$ power = current^2 × resistance

$E = P\,t$ energy transferred = power × time

Every appliance has a **power rating**. Some typical values:

phone charger	10 W	television	100 W
LED light bulb	8 W	microwave oven	800 W
laptop	60 W	kettle	2300 W
games console	150 W	electric shower	9000 W

More power means more current. Rearranging $P = V\,I$ gives $I = P \div V$. A 2300 W kettle on the 230 V mains draws $2300 \div 230 = 10$ A; the phone charger draws less than 0.1 A.

D. How this ties together — and where today goes

Every circuit you have built so far ran from a cell or a power pack. That is **direct** potential difference (voltage): it pushes the current one way only.

Plug something into the wall and you are using the **mains**. It is **alternating**, at about 230 V, and because it is dangerous the cable has three wires to keep you safe. Your home is wired in **parallel**, so every appliance gets that full 230 V.

Add up every kettle, shower and television in the country and the power needed is enormous. It comes from power stations, often hundreds of kilometres away, along cables. Those cables have resistance, so the current **heats** them: $P = I^2\,R$. The current is **squared**, so doubling the current makes the heating four times bigger.

Today follows four questions, each one leading to the next:

1. A battery gives 1.5 V. The socket gives 230 V. Where does that come from?
2. So how does it get to everyone's house?
3. 230 V is dangerous. How does the cable keep you safe?
4. Batteries give direct current. What comes out of the socket?

Question 1. A battery gives 1.5 V. The socket gives 230 V. Where does that come from?

Not from a battery. Power stations generate electricity at about **25 000 V**, and your home needs about **230 V**. On the way, the potential difference (voltage) is changed by a **transformer**.

- A **step-up transformer** *increases* the potential difference (voltage).
- A **step-down transformer** *decreases* it.

The transformer equation (Higher tier, on the equation sheet). The power going in equals the power coming out:

$$\boxed{V_p\, I_p = V_s\, I_s}$$

p.d. across primary coil × current in primary coil = p.d. across secondary coil × current in secondary coil

p is the **primary** (the side going *in*); s is the **secondary** (the side coming *out*). Each side is simply $P = V\,I$, so when the potential difference (voltage) goes **up**, the current goes **down**.

Transformers only work with **alternating** current. Hold on to that for Question 4.

Worked example — find a current

A transformer has a p.d. of 230 V and a current of 2.0 A in its primary coil. The p.d. across its secondary coil is 23 V. Calculate the current in the secondary coil.

1. Write it $V_p\, I_p = V_s\, I_s$

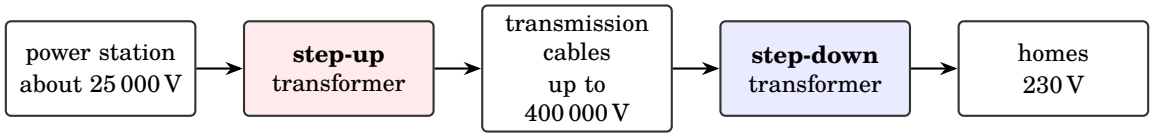
2. Substitute $230 \times 2.0 = 23 \times I_s$ so $460 = 23 \times I_s$

3. Answer $I_s = 460 \div 23 =$ **20 A**

The check. A step-*down* transformer lowers the p.d., so the current goes *up* by the same factor. Here the p.d. fell by ×10 and the current rose by ×10. Whichever side has the bigger p.d. has the smaller current.

Question 2. So how does it get to everyone's house?

Through the **National Grid**: the system of cables and transformers that links power stations to homes, schools and factories.



A **step-up** transformer raises the potential difference (voltage) from the power station to the transmission cables. A **step-down** transformer lowers it to a much safer value for use at home.

Why step up at all? It makes the Grid efficient.

1. The power station sends a fixed power, $P = V\,I$. A **higher p.d.** means a **smaller current** for the same power.
2. The cables have resistance, and they heat up by $P = I^2\,R$. A smaller current means much less energy transferred to heating the cables.
3. So less energy is wasted, and more of it reaches the consumer.

The current is *squared*, so halving the current cuts the heating to a quarter.

Question 3. 230 V is dangerous. How does the cable keep you safe?

Most appliances connect to the mains through a cable with three wires inside. Each has a colour, so you can tell them apart.

Wire	Colour	p.d.	Its job
Live	brown	about 230 V	carries the alternating p.d. from the supply
Neutral	blue	about 0 V	completes the circuit
Earth	green and yellow stripes	0 V	a safety wire; it carries a current **only when there is a fault**, and stops the appliance casing becoming live

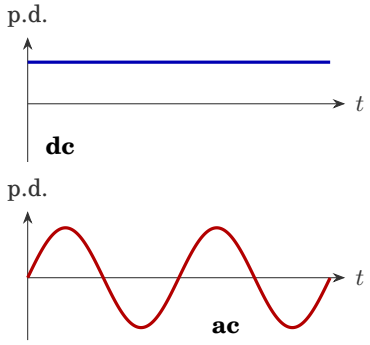
Why the live wire is dangerous. Your body is at 0 V, like the earth. Touch the live wire and there is a potential difference (voltage) of about 230 V across you, so a current flows through you to earth — an electric shock. The switch in an appliance sits on the live side, so even with the switch **off** the wire up to the switch is still at 230 V. Any connection between the live wire and earth lets a large current flow.

Question 4. Batteries give direct current. What comes out of the socket?

Direct (dc). The potential difference (voltage) stays in **one direction**, so the current flows one way only. Cells and batteries supply dc.

Alternating (ac). The potential difference (voltage) **keeps reversing direction**, so the current flows one way, then the other, over and over.

The socket gives ac: about **230 V** at a frequency of **50 Hz** — it reverses direction and back 50 times every second. The Grid runs on ac because transformers need it.



Questions

For every calculation, show the equation, the substitution and the answer with its unit.

1) **Calculate** the current in the secondary coil of a step-up transformer. The primary coil has a p.d. of 25 000 V and a current of 400 A. The secondary coil has a p.d. of 400 000 V. *Use* $V_p\, I_p = V_s\, I_s$. **[3]**

$25\,000 \times 400 = 400\,000 \times I_s$ *so* $10\,000\,000 = 400\,000 \times I_s$ *and* $I_s =$ **25 A**. *Check: p.d. up, current down.*

2) **Calculate** the current in the primary coil of a step-down transformer. The primary coil has a p.d. of 11 000 V. The secondary coil supplies 230 V and a current of 110 A. **[3]**

$11\,000 \times I_p = 230 \times 110 = 25\,300$ *so* $I_p = 25\,300 \div 11\,000 =$ **2.3 A**. *Check: the primary has the bigger p.d., so the smaller current.*

3) **Name** the type of transformer found **(a)** between a power station and the transmission cables, and **(b)** between the transmission cables and our homes. **[2]**

(a) step-up (b) step-down

4) **Explain** why the National Grid transmits electricity at a very high potential difference. **[3]**

For the same power ($P = V\,I$), a higher p.d. means a smaller current. A smaller current means less heating of the cables ($P = I^2\,R$). So less energy is wasted and the Grid is more efficient.

5) A power station sends out 100 MW (100 000 000 W). The cables to a town have a total resistance of 2.0 Ω.

(a) Calculate the current in the cables if the power is sent at 25 000 V.

$I = P \div V = 100\,000\,000 \div 25\,000 =$ **4000 A**

[2]

(b) Calculate the power transferred to heating the cables at that current.

$P = I^2\,R = 4000^2 \times 2.0 = 32\,000\,000\,W =$ **32 MW**

[2]

(c) Repeat (a) and (b) for the same power sent at 400 000 V.

$I = 100\,000\,000 \div 400\,000 = 250\,A;\ P = 250^2 \times 2.0 = 125\,000\,W =$ **0.125 MW**

[3]

(d) Use your answers to explain why step-up transformers are used.

At 400 000 V the heating loss falls from 32 MW to 0.125 MW, so far less energy is wasted: stepping up the p.d. makes transmission much more efficient.

[2]

6) Complete the table for the three wires in a three-core cable.

[3]

Wire	**Colour**	**Its job**
Live	*brown*	*carries the alternating p.d. from the supply*
Neutral	*blue*	*completes the circuit*
Earth	*green and yellow stripes*	*safety wire; carries a current only if there is a fault*

7) **Explain** why touching the live wire of a lamp can give an electric shock, even when the lamp is switched off. **[3]**

The switch is on the live side, so the live wire is still at about 230 V. The person is at 0 V (earth), so there is a p.d. across them. A current flows through them to earth.

8) **State** the potential difference and the frequency of the UK mains supply. **[2]**

about **230 V** *and* **50 Hz**

9) **Describe** the difference between a direct potential difference and an alternating potential difference. **[2]**

Direct: stays in one direction (current flows one way only). Alternating: keeps reversing direction (current flows one way, then the other).